

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's break down the problem step by step to determine the total time it takes for Carla to download the file .

1 . **Calculate the normal download time without any interruptions : **

- The total size of the file is 200 GB .
- Carla's normal download rate is 2 GB / minute .
- To find the total time required to download 200 GB at this rate :

$$\text{Total time} = \frac{\text{Total size}}{\text{Download rate}} = \frac{200 \text{ GB}}{2 \text{ GB/minute}} = 100 \text{ minutes}$$

2 . **Determine the point where Windows forces a restart : **

- 40 % of the way through the download :

$$40\% \text{ of } 200 \text{ GB} = 0.40 \times 200 = 80 \text{ GB}$$

- The time taken to download 80 GB at 2 GB / minute :

$$\text{Time to download } 80 \text{ GB} = \frac{80 \text{ GB}}{2 \text{ GB/minute}} = 40 \text{ minutes}$$

3 . **Account for the forced restart and re-download : **

- The restart takes 20 minutes .
- After the restart , Carla needs to re-download the entire 200 GB from the beginning .

4 . **Calculate the time to re-download the entire file : **

- As calculated earlier , the total time to download 200 GB is 100 minutes .

5 . **Sum up all the time segments : **

- Time to download the first 80 GB : 40 minutes
- Time for the restart : 20 minutes
- Time to re-download the entire 200 GB : 100 minutes
- Total time :

$$\text{Total time} = 40 \text{ minutes} + 20 \text{ minutes} + 100 \text{ minutes} = 160 \text{ minutes}$$

Thus , the total time it takes for Carla to download the file , including the restart , is 160 minutes .

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1 . Normal download time for 200 GB : 100 minutes
2 . Time to download 80 GB : 40 minutes
3 . Restart time : 20 minutes
4 . Re-download time for 200 GB : 100 minutes
5 . Total time : 40 + 20 + 100 = 160 minutes
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<answer> 160 </answer>